SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I Kaviya · G (192524281) (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kaviya · G

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

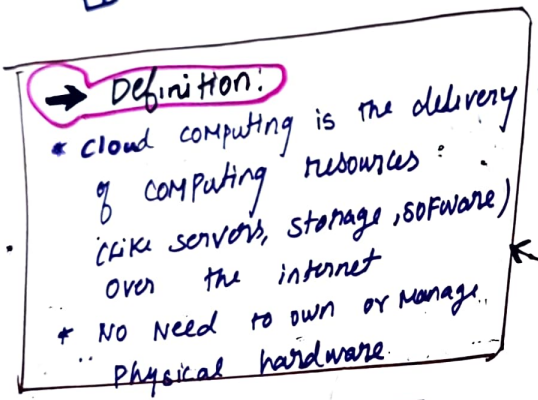
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

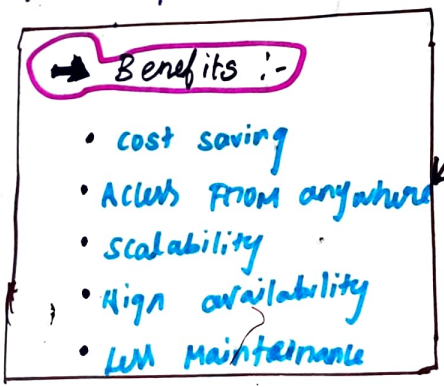
Total Marks:

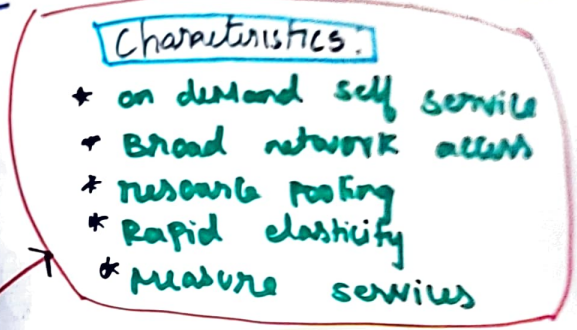
KAVIYA G
192524291
B Tech AIDS

CONCEPT MAP -

1. Cloud computing fundamentals

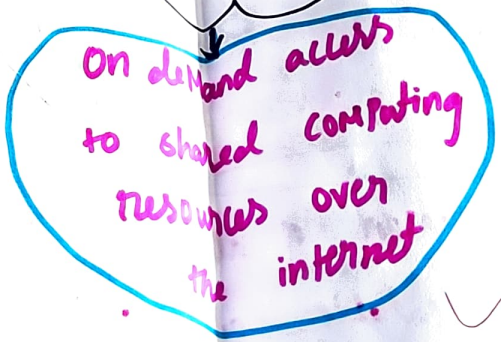


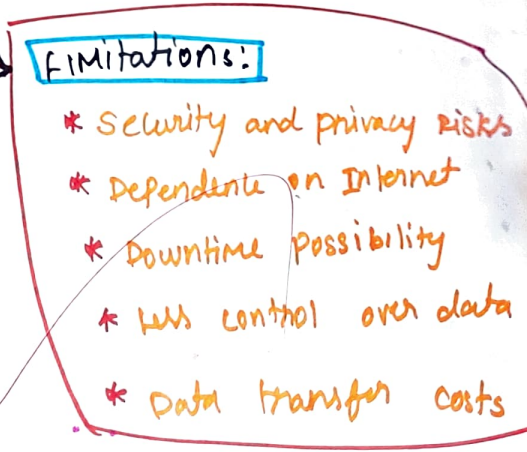




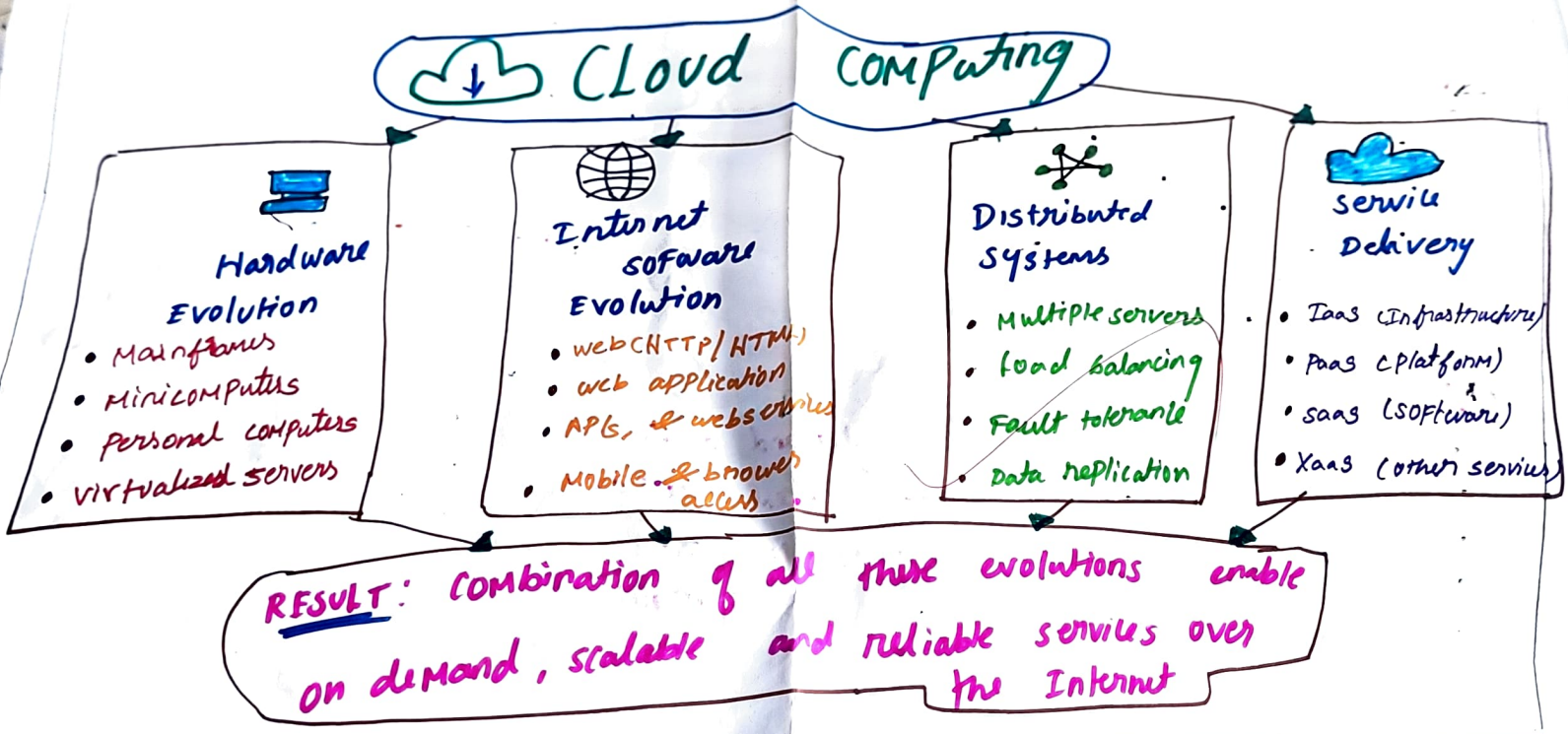
Flexible - Accessible - Efficient

CLOUD COMPUTING





2. EVOLUTION CLOUD COMPUTING



3. SERVICE MODELS

Iaas

(Infrastructure as a service)

Provider :-
- Physical hardware
- virtualization
- Networking
- Storage
- Data centers

User :-
- OS, applications
- Data, security
- configuration

Eg : AWS, EC2, Azure, VMs

Paas

(Platform as a service)

Provider :-
- Runtime environment
- Middleware
- Development tools
- Infrastructure

User :-
- Application
- Data
- Manage code

Eg : Heroku, google APP Engine, Azure App Service

Saas

(software as a service)

Provider
- Complete application
- Updates, Maintenance
- Security
- Availability

User
- Use the software
- Manage data
- User setting

Eg : gmail, Microsoft 365

Xaas

(Anything as a service)

Provide
- Specific service
- Management Security
- Availability

User
- Use the service
- Manage data
- Follow service Rule

Eg : DBaas, Faas STaas, Ca

4) Virtualization and

...vice as Enablers of Cloud

virtualization

- Create multiple virtual machines
- shares physical resources
- Improves resource utilization
- Provides isolation & security

web services :

- Uses standard protocols (HTTP / REST)
- Enable application communication
- Supports interoperability
- Provides APIs

Enable ⚙ Cloud

Supports cloud Deployment

- Lets multiple VMs run on one server
- Easy to provision and manage
- Improve reliability and security

Enables scalability

- Quickly add or remove resources
- Handle varying user demand
- Make efficient use of resource

14/10/26